Brainstorming & Idea Prioritization

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

1. Objective

Classroom attendance and attentiveness are still handled by manual roll calls and the lecturer's intuition. This session explored ways AI could automate both, while staying affordable and respecting student privacy. Ideas were generated freely first, then scored and prioritized.

2. Brainstorming Rules Followed

- Quantity over quality in the first round; no idea is rejected while generating.
- Every idea is written as a single sentence describing who benefits and how.
- Ideas are grouped into themes, then scored against agreed criteria.
- Constraint kept in mind throughout: it must run on a standard laptop, offline, with student consent.

3. Ideas Generated (Themes)

#	Theme	Idea
1	Attendance	Face-recognition roll call that marks students automatically as they sit down.
2	Attendance	RFID / ID-card tap-in at the door.
3	Attendance	QR code on the projector that students scan with phones.
4	Attendance	Presence-over-time attendance: credit is earned by how long a student is visibly present, not a single snapshot.
5	Engagement	Estimate attention from head pose and eye gaze using the camera.
6	Engagement	Detect classroom noise and disturbances from the microphone.
7	Engagement	Fuse vision and audio into one engagement score for the whole class.
8	Engagement	Student-facing pop-up quizzes to measure attention.
9	Teaching support	Rule-based teaching suggestions shown live (e.g. ask a check-in question when attention dips).
10	Teaching support	Cloud LLM that summarizes each lecture.
11	Privacy	Run everything locally; never upload video or biometric data.
12	Privacy	Consent-first enrollment; unenrolled people appear only as anonymous track IDs.
13	Reporting	Dashboard with live monitoring, history, and CSV export.
14	Reporting	Manual override so the lecturer can correct mistakes with an audit reason.

4. Prioritization Criteria

Each idea was scored from 1 (low) to 5 (high) on four criteria. Total = sum of the four (max 20).

- **Impact** - how strongly it solves attendance or attentiveness problems.
- **Feasibility** - can a first-year team build it with open-source Python tools?
- **Privacy fit** - does it avoid cloud upload and respect consent?
- **Cost** - 5 means it needs no extra hardware or paid services.

#	Idea (short)	Impact	Feasib.	Privacy	Cost	Total	Decision
1	Face-recognition roll call	5	4	3	5	17	Build
2	RFID tap-in	3	3	5	2	13	Drop (proxy risk, hardware)
3	QR code scan	3	5	4	5	17	Drop (easy to share, proxy risk)
4	Presence-over-time attendance	5	4	4	5	18	Build (core)
5	Head pose + gaze attention	5	4	4	5	18	Build (core)
6	Audio disturbance detection	4	4	5	5	18	Build
7	Fused engagement score	5	3	4	5	17	Build (core)
8	Student pop-up quizzes	3	3	5	5	16	Later
9	Rule-based teaching tips	4	5	5	5	19	Build
10	Cloud LLM summaries	3	3	1	2	9	Drop (breaks offline goal)
11	100% local processing	5	5	5	5	20	Build (principle)
12	Consent-first enrollment	4	5	5	5	19	Build
13	Dashboard + CSV export	4	5	5	5	19	Build
14	Manual override + audit	4	5	5	5	19	Build

Scores are the team's judgement during the session, used to rank ideas rather than to measure them precisely.

5. Final Selection

The ideas with the highest combined value were merged into a single product: an offline, privacy-preserving agent that **(a)** records attendance by verified presence time, **(b)** computes a fused vision + audio engagement index, **(c)** issues rule-based teaching recommendations, and **(d)** stores everything locally in SQLite with a Streamlit dashboard.

- **Must have:** face detection and tracking, consent-based recognition, head pose/gaze, audio classification, fusion, attendance, dashboard.
- **Should have:** recommendation engine, manual override, CSV export, session history.
- **Later:** multi-camera support, time-series storage for per-frame metrics, local language model summaries, student consent portal.